

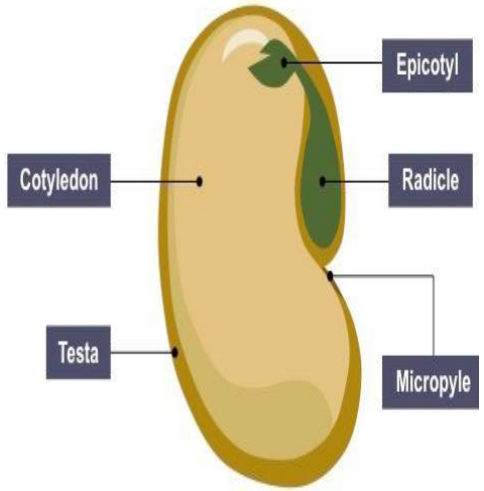

SEED

Seed

Seed is a fertilized matured ovule together covered with seed coat is called seed or it is a propagating material.



- A seed is an embryonic plant enclosed in a protective outer covering.
- The formation of the seed completes the process of reproduction in seed plants.



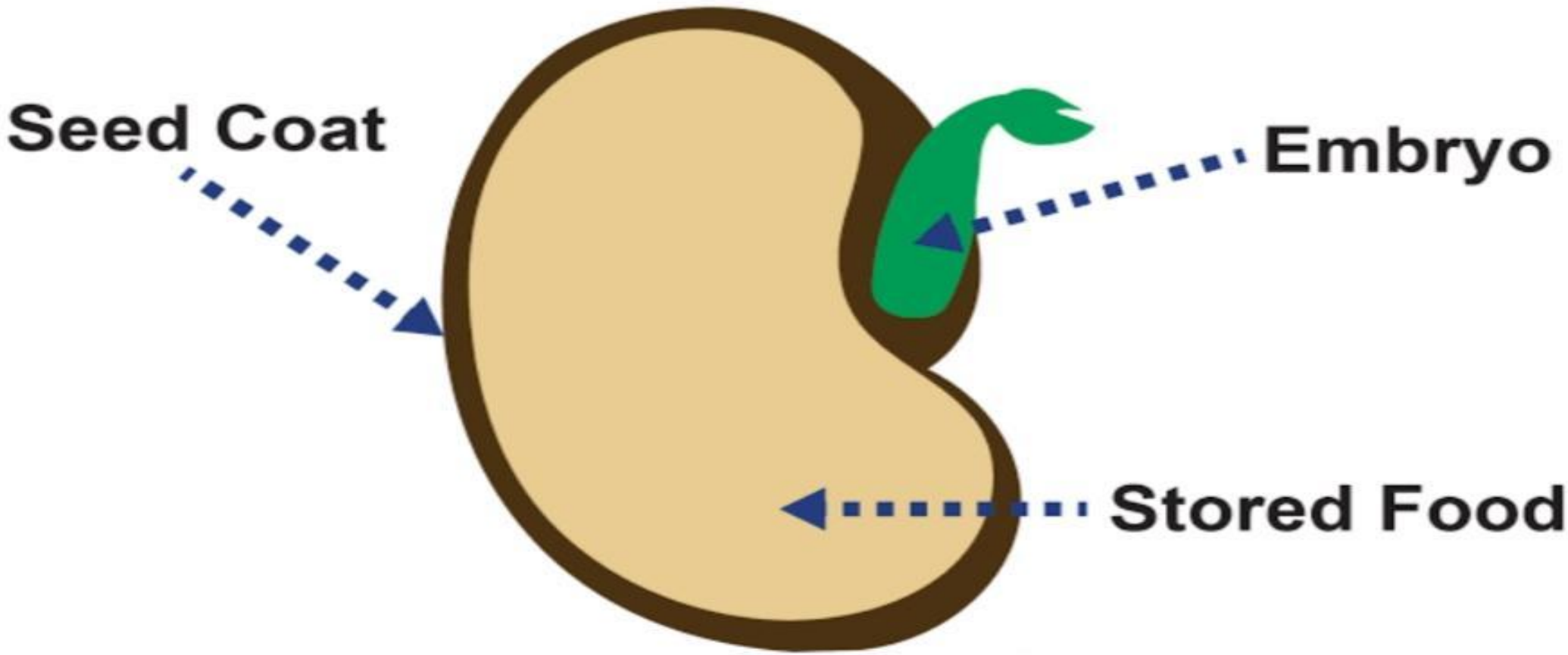


Seed Structure

A typical seed includes three basic parts:

1. Embryo
2. Endosperm
3. Seed Coat

Parts of a Seed





Characteristic of good seed

- It must be true to its type.
- The seeds must be healthy, pure and free from all the inert materials and weed seeds.
- The seeds must be viable, the germination capacity is up to the standard and it has been tested recently
- The seeds must be uniform in texture and structure.
- The seed must not be affected by any seed born disease.



Types of seed

- **Based on the number of cotyledons:-**

1. Monocotyledonous seed:

These are the seeds with only one cotyledon.
Examples: wheat, corn, rice, etc.

- **Germination usually hypogeal (except in Onion)**
- **Roots are advantitious.**
- **Secondry growth absent.**
- **Usually Endospermic.**

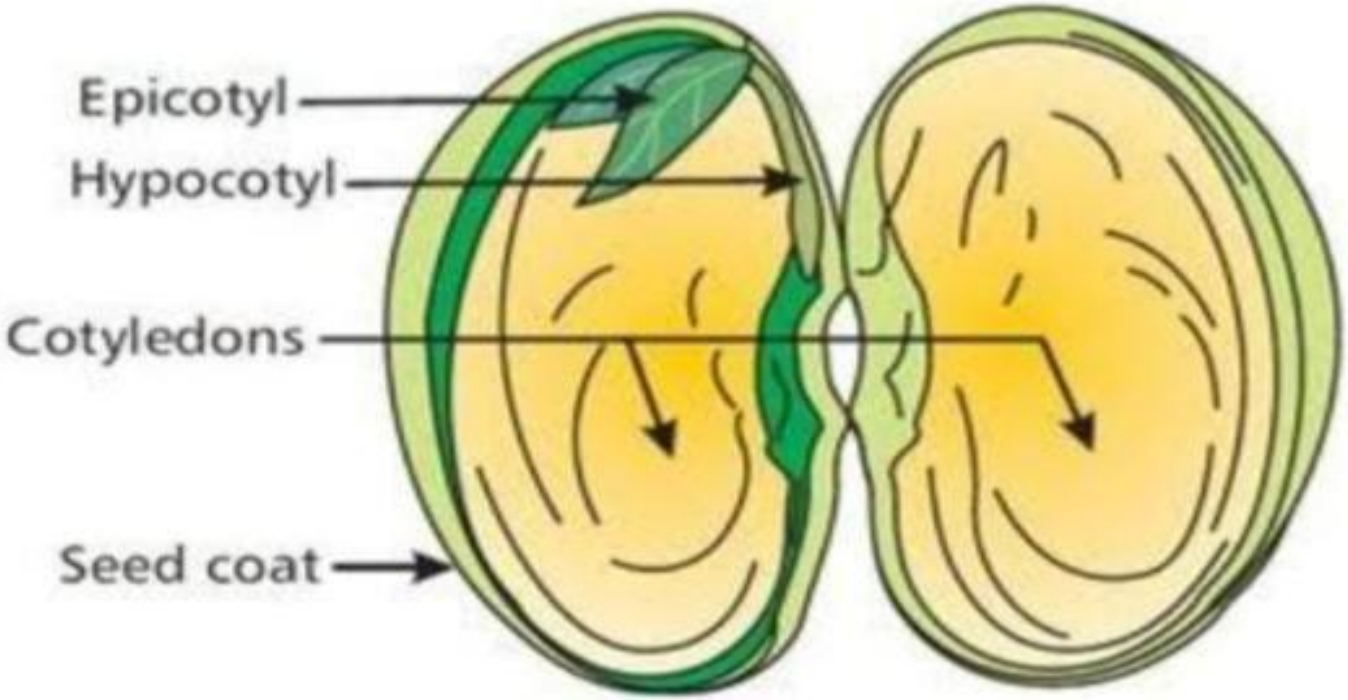


2. Dicotyledonous seed:

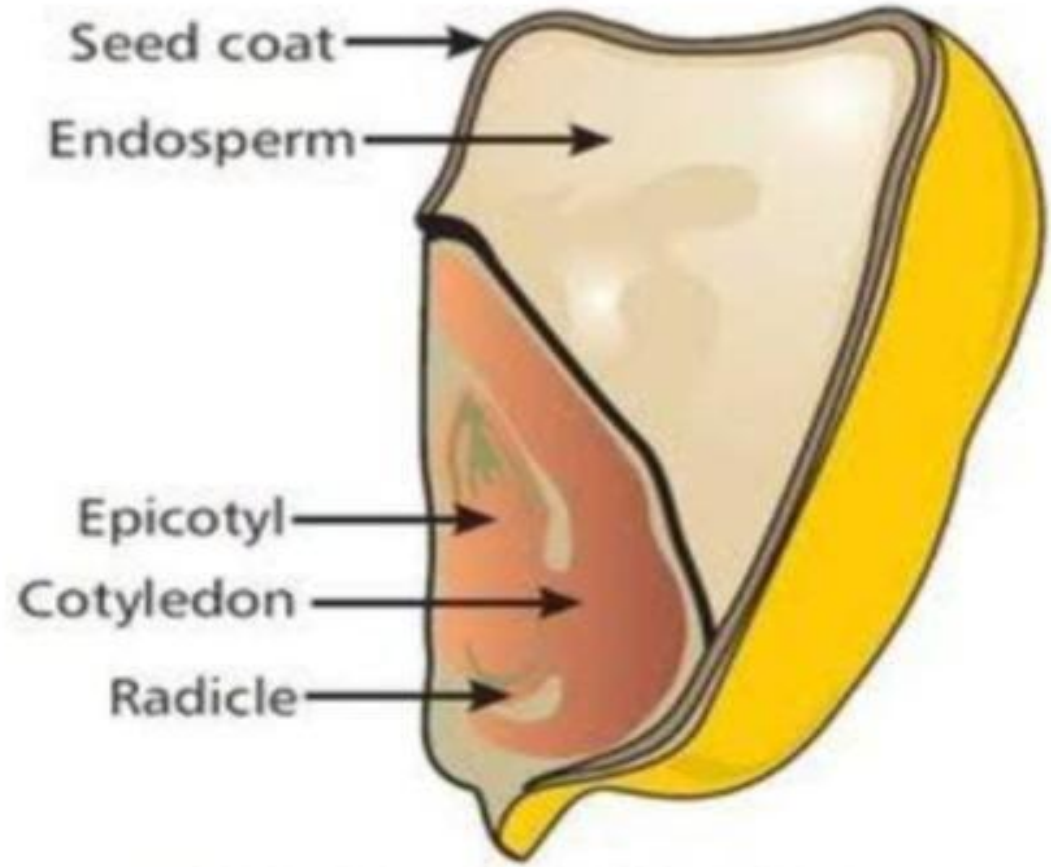
These are the seeds that have two cotyledons.
Examples: pea, bean, mango, etc.

- Germination usually epigeal (except in pea and gram)
- Usually non Endospermic.
- Secondry growth often present.
- Flower parts in multiples of four or five

Monocot vs Dicot seed



Dicotyledon



Monocotyledon



- **Based on the presence or absence of endosperm:-**

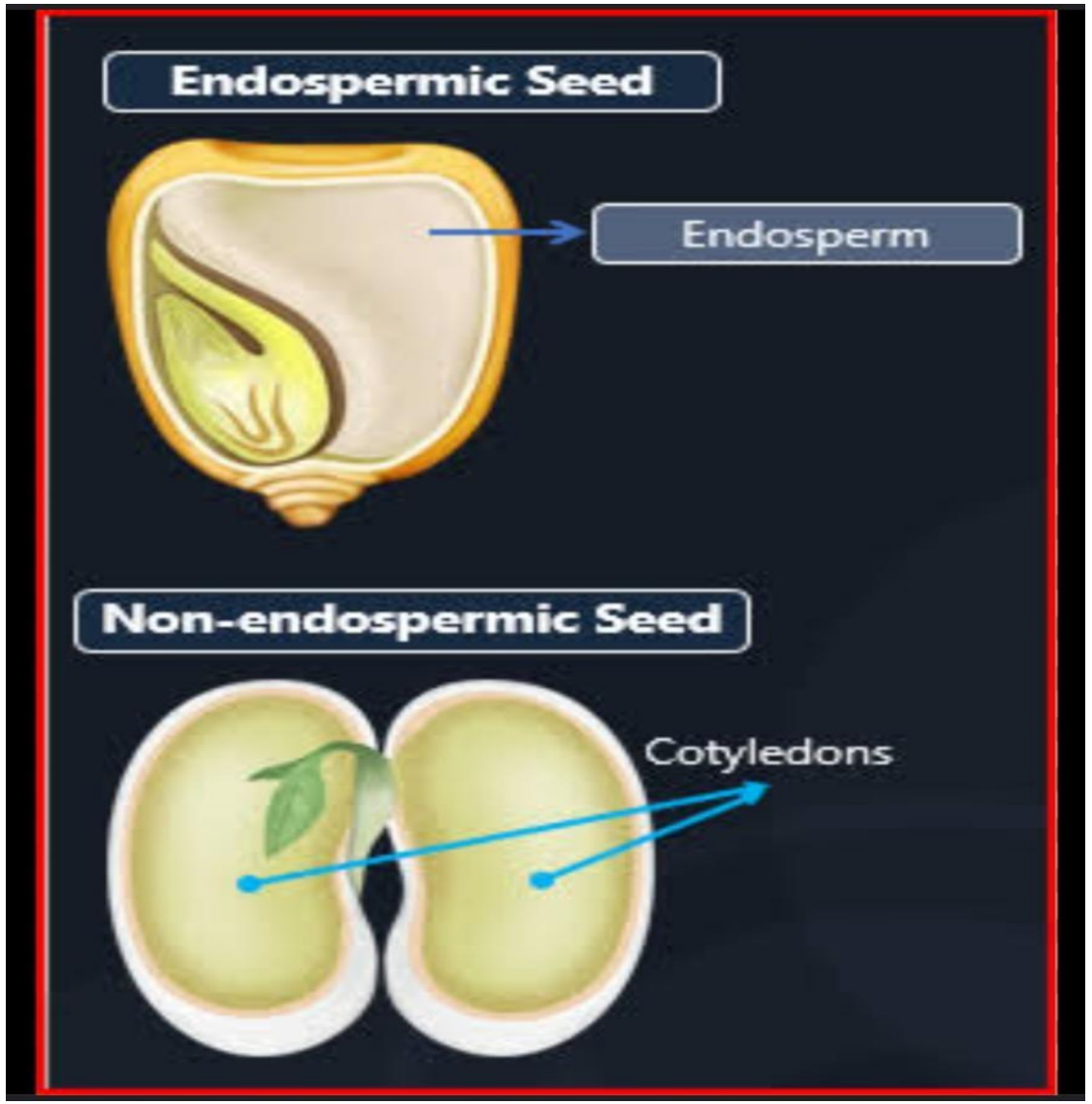
1. Albuminous seeds (Endospermic):

Seeds with endosperm are called endospermic seeds.
Ex: Castor, poppy, custard apple, etc.

2. Exalbuminous seeds (Non-endospermic):

Seeds without endosperm are called non-endospermic seeds.
Ex: Gram, pea, bean, etc.




Endospermic Seed
Endosperm
Non-endospermic Seed
Cotyledons



- **Based on the storage:-**

1.Orthodox seed:

Orthodox seeds are seeds which will survive drying and/or freezing during ex-situ conservation.
Ex: Capsicum annum, Citrus aurantifolia, mostly grain and legumes

- **Can be stored for longer period.**
- **Dried to moisture content of 10% or less and successfully stored at sub freezing temperatures**



2.Recalcitrant seed:

- Seed which cannot survive desiccation when dry to low moisture content
- Cannot be stored for the longer period
- Sensitive to chilling or freezing temperature
- The moisture content of recalcitrant seeds at maturity is in the range of 30 to 70 percent

Ex: Coffee, Mango, Rubber, Neem, Jackfruit, etc



- **Based on the type of food we get from seeds:-**



1.Grains: Grains include rice, wheat, jowar, maize and ragi.



2.Pulses: Pulses include grams, peas and beans.



3.Oilseeds: Oilseeds include groundnut, sesame, mustard, coconut and sunflower seed.



Reference

https://www.toppr.com/guides/biology/morphology-of-flowering-plants/the-seed/

Plant Breeding: Princples and Methods : B D Singh

THANK YOU